

Vansh Mahajan

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EDUCATION

B.Tech. in Computer Science
and Engineering

**Vellore Institute of Technology,
Vellore, India** 2022-2026
(ongoing)

CGPA: 8.75

COURSE CERTIFICATIONS

- MERN Full Stack Certification
– ETHNUS
(Credential ID: KGRJY8W5)

- Oracle Cloud Infrastructure
2025 AI Foundations
– Oracle
(Credential ID: OC5851383)

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Jul 2025 Mar 2025 Mar 2025

Leadership & Activities

Soft Computing
Research Society– VIT
Vellore Core Committee
Member

PROFILE

As a Software Engineer with strong foundations in programming, full-stack web development, and data engineering, I have developed strong foundations in my domain, especially ML and DL, have some experience in applying them to real-world projects, and am eager to contribute in more projects, learning more throughout the journey.

WORK EXPERIENCE

Web Development and Hosting Intern – CT Software Solutions (May 2025– Jul 2025)

- Contributed to development and optimization of Fitness360 web admin panel
- Optimized SQL queries and improved UI interactivity using Laravel, AJAX and DataTables
- Built five Highcharts dashboards to monitor user engagement and fitness progress
- Improved UI/UX using responsive Bootstrap design
- Collaborated on testing and deployment using Git

Data Engineer Intern – KalkiFI AI Solutions Pvt. Ltd. (Mar 2024– Aug 2024)

- Processed structured datasets using Python, Pandas and NumPy
- Built ETL pipelines for consistent and reliable data workflows
- Optimized SQL queries and database workflows
- Developed analytical dashboards using Power BI and Matplotlib
- Collaborated with teams using Git

SKILLS

- **Programming Languages:** C/C++, C#, Java, JavaScript, Python, PHP, Go, Kotlin
- **Frameworks & Libraries:** React, Node.js, Express.js, Flutter, Laravel, TensorFlow, Highcharts, Bootstrap, Tailwind, MERN fullstack, spring boot, Flask
- **Databases & Data Management:** SQL, PLSQL, NoSQL, MongoDB, DBMS
- **DevOps & Tools:** AWS, Docker, Git, GitHub, REST, Ajax, Kubernetes
- **Concepts & Methodologies:** Agile, OOP, API, Data Structures, AI/ML, Machine Learning, Infrastructure As Code, Software Development, Programming Knowledge, Code Optimization, Debugging, Automation, Results Driven

PROJECTS

Real-Time 360° Assistive Vision System

- Created a wearable 360° assistive vision headset integrating YOLOv12, LiDAR fusion, and DCP dehazing for dependable real-time hazard detection.
- Engineered a risk scoring model to assess object proximity, class, and motion for intelligent prioritization of threats.
- Built a real-time audio feedback system to deliver intuitive guidance for visually impaired users and individuals in safety-critical environments.

Intrusion Detection System using XGBoost (Cloud Security)

- Developed a machine learning-based Intrusion Detection System (IDS) for cloud environments using XGBoost, Random Forest, and Neural Networks to detect multiple cyberattack types.
- Implemented end-to-end pipeline including data preprocessing, feature engineering, and multi-class classification on the KDD dataset, achieving ~99.9% accuracy and high ROC-AUC.
- Leveraged AWS services (S3, SageMaker) for scalable training and deployment, while optimizing model performance to reduce false positives and improve detection of rare attacks.

Resume Screening RAG Pipeline

- Developed an LLM-based chatbot to assist hiring managers in resume screening using RAG and RAG Fusion techniques.
- The system uses hybrid retrieval to match resumes with job descriptions.
- Built with LangChain, OpenAI, HuggingFace, FAISS, and Streamlit.

License Plate Recognition System

- Built automated vehicle authorization system using Python, OpenCV and OCR.
- Developed image processing pipeline for license plate detection.
- Integrated SQLite database for real-time verification.
- Improved detection accuracy through testing.

Full Stack Development (Web Development Project) – Real-time Chat Application

- Built a full-stack web chat application enabling private messaging between users.
- Implemented real-time communication using Socket.io, RESTful APIs with Express, MongoDB database, JWT authentication, and bcrypt for password hashing.
- Frontend developed with React, React Bootstrap, and React Router.
- Includes user registration, login, chat interface, and notification system.